

# Scaling TinyML

Linh B. Ngo

West Chester University of Pennsylvania

CRA / NAIRR Pilot Classroom Expansion Conference  
Research-Emerging Institutions with Large Computing Programs

June 2026



**Classroom  
Expansion  
Conferences**

## What is scaling?

- ▶ Student demand for AI, machine learning, and applied computing.
- ▶ Expectations from industry, graduate programs, and interdisciplinary partners.
- ▶ Tool churn: models, libraries, cloud platforms, boards, APIs.

## What is not scaling?

- ▶ Faculty preparation time.
- ▶ Local computing support.
- ▶ Stable hardware/software stacks.
- ▶ Departmental bandwidth for new courses.
- ▶ Baseline preparation/prerequisites for students

## Original Course Description

- ▶ Harvard's TinyML
- ▶ A graduate/applied course in machine learning for embedded devices
  - Inexpensive microcontrollers
  - Local tooling
  - Deployable TinyML workflows

## Why this works

- ▶ Requires minimal new prep and technical overhead.
- ▶ Demonstrates a full AI pipeline at classroom scale.
- ▶ Exposes real resource constraints early.
- ▶ Is already decomposed into reusable modules.
- ▶ Has demonstrated scaling capability (MOOC)

## Practical barriers

- ▶ Ready-made lecture materials
  - Dependencies are loosely related and do not line up well with institutional schedules
- ▶ Relaxed prerequisites to support MOOC
  - Fundamentals of embedded systems are omitted, especially regarding sensor capabilities.
  - Too much emphasis on various workflows (magic wand, voice detection, anomaly detection)
- ▶ Read-made hardware kit
  - Out of stock, cannot support the strict institutional timeline
- ▶ Outdated libraries and IDEs
  - Harder to transition toward research activities

# Adaptation principle 1: Stable environment

## ► Hardware

- Simpler hardware: single board instead of complex kit
- Remove camera; gain proximity, humidity, and temperature sensors
- Cheaper and readily available: Arduino Nano 33 BLE Rev2

## ► Software

- VSCode + PlatformIO instead of Arduino IDE
- Ready-made starter repository
- Cross-platform setup



Starter repository

## Adaptation principle 2: Teach the complete workflow at smaller scale

### Full workflow

- ① Capture data.
- ② Label and clean data.
- ③ Train a model.
- ④ Deploy to an edge device.
- ⑤ Observe the real system.

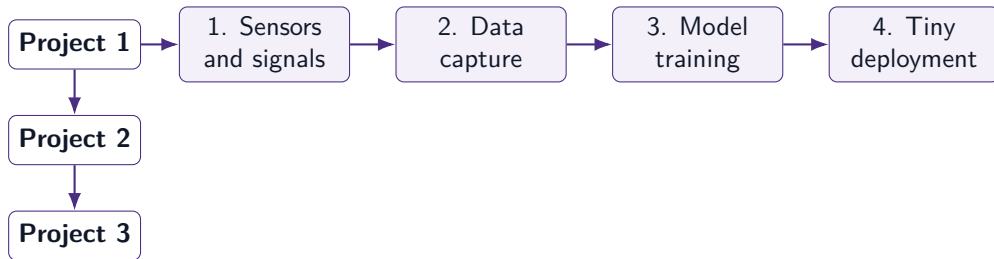
### Practical constraints

- ▶ Isolate sensors under limited conditions and modest models.
- ▶ Emphasize reproducible notebooks and scripts over polished demos.
- ▶ Let students see quantization, latency, memory limits, and drift.
- ▶ Reserve large compute for later stages.

### Takeaway

Students learn AI better when the system is small enough that its failures are visible.

## Adaptation principle 3: Extend the modular structure



### Modular structure

- ▶ Full workflow per project (versus multiple project stage per workflow stage)
- ▶ Helps students to internalize the big picture

### Workflow emphasis

- ▶ In-depth discussion at each stage of the workflow.
- ▶ Helps students understand the technical details (e.g: why do we use this sensor)

## Adaptation principle 4: Eye toward the future

- ▶ Can students from related disciplines take this course
  - WCU: Collaboration with Biomedical Engineering for a new AI Engineering certificate
- ▶ Can individual modules from this course be lightened at the trade-off of prerequisites?
  - Computer Systems/Organizations, System Programming, Embedded Systems
  - Intro to ML, Applied ML
  - Data Science



- ▶ Simpler, more structured TinyML materials help prepare students for research, even when they haven't taken the course yet.
- ▶ First batch of student projects
  - Toward digital shadow/digital twins with Arduino Nano 33 BLE Rev2 and Webot
  - LLM-based navigation reasoning with on-board sensors
- ▶ Robotic Club?

# Potential improvements with NAIRR resources

## Compute

- ▶ Better/faster model training via ACCESS resources
- ▶ FABRIC/Qemu for simulated large-scale deployment

## Data

- ▶ Additional data sources/sensors via Sage Grande Testbed
  - Reduce direct hardware interaction
  - Improve data collection, model selection, creation, and training

The NSF describes NAIRR as infrastructure for access to computing, software, data, models, educational resources, and expertise; the CRA Classroom Expansion effort emphasizes capacity building and communities of practice.

## Question?



Lecture nodes (Will be updated continuously)



Starter repository

## Selected sources and alignment notes

- ▶ U.S. National Science Foundation. *National Artificial Intelligence Research Resource*. NSF describes NAIRR as national infrastructure for computing, software, data, models, educational resources, and expertise.  
<https://www.nsf.gov/focus-areas/ai/nairr>
- ▶ Warden, P. and Situnayake, D. *TinyML: Machine Learning with TensorFlow Lite on Arduino and Ultra-Low-Power Microcontrollers*. O'Reilly, 2019.
- ▶ Reddi, V. J. et al. "Widening Access to TinyML Network by Establishing Best Practices in Education." 2021.